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Spectral algebraic

geometry I

Last time. More on the functor of points, quotients

Today. Begin Spectral algebraic geometry

Summary from last time

(1) The functor of points of a scheme satisfies fpqc descent

(2) We can define $\mathrm{QCoh}(-)^{\heartsuit} : \mathrm{Fun}(\mathrm{CAlg}, \mathrm{Set})^{\heartsuit} \rightarrow \mathrm{CMon}(\mathrm{Cat}_1)$
by extending $\mathrm{Mod}_{(-)}^{\heartsuit} : \mathrm{CAlg} \rightarrow \mathrm{CMon}(\mathrm{Cat}_1)$

$$\mathrm{QCoh}(X)^{\heartsuit} := \lim_{\mathrm{Spec}(A) \rightarrow X} \mathrm{Mod}_A^{\heartsuit}$$

(3) $\mathrm{QCoh}(-)^{\heartsuit}$ satisfies fpqc descent

Origin of the theory of stacks. If we're given a scheme X with an action of a group scheme G , the quotient X/G in $\mathrm{Fun}_{\mathrm{fpqc}}(\mathrm{CAlg}, \mathrm{Set})$ may not be very interesting / not capture equivariant geometry.

- > If $X = *$, then the quotient is always $*$
- > The image of $\mathbb{A}^{n+1}/G_m$ at the origin should be interesting, but isn't
- > Basically we want an analogue of the topologist's BG that classifies G -bundles
 - The problem is that while the groupoid of G -bundles forms a sheaf, the set of isomorphism classes of G -bundles does not
 - So we should enlarge our world to $\text{Fun}_{\text{fpgc}}(\text{CAlg}, \text{Gpd})$
 Note that $\text{Gpd} \simeq \text{An}_{\leq 1}$
- > But the problem persists. We should enlarge to $\text{Fun}_{\text{fpgc}}(\text{CAlg}, \text{An}_{\leq 2})$, then $\text{Fun}_{\text{fpgc}}(\text{CAlg}, \text{An}_{\leq 3})$, ..., then $\text{Fun}_{\text{fpgc}}(\text{CAlg}, \text{An})$

Spectral algebraic geometry beginnings

So far. We have the symmetric monoidal stable ∞ -category $(\mathcal{S}_p, \otimes, \mathbb{S})$.

> It is also reasonable to call this $\text{Mod}_{\mathbb{S}}$

> We have the free $\mathbb{S}$ -module functor

$$\mathbb{S}\langle - \rangle := \sum_{+}^{\infty} : (A_n, \wedge) \rightarrow (\text{Mod}_{\mathbb{S}}, \otimes)$$

which is symmetric monoidal.

> We can talk about commutative algebras in $\text{Mod}_{\mathbb{S}}$, as well as commutative coalgebras.

- This is actually more technical than the case of commutative monoids with respect to cartesian product.

- We won't go into the details, but you can assume it works as expected.

- > So to formulate non-affine toric geometry, we need an analogue of scheme theory
- Before doing so, let us explain the relationship between classical ring theory and commutative algebras in $\mathcal{S}p$.

Comparison to Classical algebra

Thm. The unique colimit-preserving functor $Sp \rightarrow D(\mathbb{Z})$ determined by $S \mapsto \mathbb{Z}$ refines to a symmetric monoidal functor

$$(Sp, \otimes) \longrightarrow (D(\mathbb{Z}), \otimes_{\mathbb{Z}}^{\mathbb{L}})$$

We denote its right adjoint by $H: D(\mathbb{Z}) \rightarrow Sp$ and call this the Eilenberg-MacLane functor.

Remark. The name is because $H\mathbb{Z}$ is the spectrum representing homology with coefficients in $\mathbb{Z}$.

Intuitionistic definition. If $\mathcal{C}$ and $\mathcal{D}$ are sym mon ∞ -categories, there is a weakening of the notion of a symmetric mon. functor $F: \mathcal{C} \rightarrow \mathcal{D}$ called a lax symmetric monoidal functor, which specifies a map

$$1_{\mathcal{D}} \longrightarrow P(1_{\mathcal{C}})$$

and maps

$$F(X_1) \otimes_D \cdots \otimes_D F(X_n) \longrightarrow F(X_1 \otimes_e \cdots \otimes_e X_n)$$

subject to all higher coherences

Fact. If $F: \mathcal{C} \longrightarrow \mathcal{D}$ is a lax symmetric monoidal functor, then F induces a functor $\text{CAlg}(\mathcal{C}) \longrightarrow \text{CAlg}(\mathcal{D})$ fitting into a commutative square

$$\begin{array}{ccc} \text{CAlg}(\mathcal{C}) & \longrightarrow & \text{CAlg}(\mathcal{D}) \\ \text{forget} \downarrow & & \downarrow \text{forget} \\ \mathcal{C} & \xrightarrow{F} & \mathcal{D} \end{array}$$

Fact. If $F: \mathcal{C} \rightleftarrows \mathcal{D}: G$ is an adjunction, then a Sym. monoidal structure on the left adjoint F induces a lax Symmetric monoidal structure on the right adjoint G .

Upshot The Eilenberg-MacLane functor $H: D(\mathbb{Z}) \rightarrow Sp$ is lax sym mon and induces a functor

$$H: \text{CAlg}(D(\mathbb{Z})) \rightarrow \text{CAlg}(Sp)$$

Observe The inclusion $\text{Mod}_{\mathbb{Z}}^{\heartsuit} \hookrightarrow D(\mathbb{Z})$ as the complexes concentrated in degree 0 is lax symmetric monoidal. This is because

$$H_0(M_1[0] \otimes_{\mathbb{Z}} \cdots \otimes_{\mathbb{Z}} M_n[0]) \cong M_1 \otimes_{\mathbb{Z}} \cdots \otimes_{\mathbb{Z}} M_n$$

> There's a natural map from a connective chain complex to its H_0 , which gives the desired map

$$M_1[0] \otimes_{\mathbb{Z}} \cdots \otimes_{\mathbb{Z}} M_n[0] \longrightarrow M_1 \otimes_{\mathbb{Z}} \cdots \otimes_{\mathbb{Z}} M_n$$

> Hence we obtain functors

$$\left\{ \begin{array}{c} \text{comm.} \\ \text{ring} \end{array} \right\} = \text{CAlg}(\text{Mod}_{\mathbb{Z}}^{\heartsuit}) \hookrightarrow \text{CAlg}(D(\mathbb{Z})) \xrightarrow{H} \text{CAlg}(Sp)$$

⚠ If A and B are ordinary commutative rings, then there is a map

$$H_A \otimes_S H_B \longrightarrow H(A \otimes_{\mathbb{Z}} B)$$

that induces an isomorphism on π_0 , but generally $H_A \otimes_S H_B$ has higher homotopy.

Note. If $\mathcal{C}$ is a sym mon ∞ -category and $A \in \text{CAlg}(\mathcal{C})$, one can talk about the ∞ -category $\text{Mod}_A(\mathcal{C})$ of (left) A -modules. This comes equipped with a forgetful functor $\text{Mod}_A(\mathcal{C}) \rightarrow \mathcal{C}$ and a left adjoint

$$A \otimes_{\mathcal{C}} (-) : \mathcal{C} \longrightarrow \text{Mod}_A(\mathcal{C}).$$

Thm. If A is an ordinary commutative ring, then the unique colimit-preserving functor $\text{Sp} \rightarrow D(A)$ given by $S_1 \mapsto A$ induces an equivalence

$$\text{Mod}_{H_A}(\text{Sp}) \xrightarrow{\sim} D(A).$$

> Moreover, this equivalence refines to an equivalence of symmetric monoidal ∞ -categories.

Convention For an ordinary ring A , we simply write A for $HA \in \text{CAlg}(\text{Sp})$, and Mod_A for

$$\text{Mod}_{HA}(\text{Sp}) \cong D(A).$$

The graded ring structure on homotopy groups

Def. A commutative algebra in spectra is called an $\mathbb{E}_\infty$ -ring spectrum or $\mathbb{E}_\infty$ -ring. An $\mathbb{E}_\infty$ -ring A is connective if the underlying spectrum of A is connective, i.e., $\pi_n(A) = 0$ for $n < 0$.

Notation. From now on, we write $\mathbf{CAlg} = \mathbf{CAlg}(\mathcal{S})$ and we write $\mathbf{CAlg}^\heartsuit = \mathbf{CAlg}(\mathbf{Mod}_\mathbb{Z}^\heartsuit)$ for the ordinary category of commutative rings. ↙ should be bold

> We write $\mathbf{CAlg}^{\text{cn}} \subset \mathbf{CAlg}$ for the full subcategory spanned by the connective $\mathbb{E}_\infty$ -rings.

Observe. Let E be a spectrum. Then since $\mathcal{S} = \sum_+^\infty *$, we see that

$$\begin{aligned} \mathrm{Map}_{\mathcal{S}}(\mathcal{S}, E) &\cong \mathrm{Map}_{\mathcal{A}n}(*, \Omega^\infty E) \\ &\cong \Omega^\infty E. \end{aligned}$$

> In particular, $\pi_0 \text{Map}_{\text{Sp}}(S, E) \simeq \pi_0(E)$ as we defined before

> Similarly,

$$\begin{aligned} \text{Map}_{\text{Sp}}(S[n], E) &\simeq \text{Map}(S, E[-n]) \\ &\simeq \Omega^\infty E[-n] \end{aligned}$$

This is the anima we earlier denoted E_{-n} . So we see that $\pi_n(E)$ as defined before is $\pi_0 \text{Map}_{\text{Sp}}(S[n], E)$

Fact. We have $\pi_0(S) \simeq \mathbb{Z}$. This follows from the unstable computation $\pi_n(S^n) \simeq \mathbb{Z}$.

Topological exercise. For $m, n \geq 0$, the map of topological spaces

$$S^{m+n} \simeq S^m \wedge S^n \xrightarrow{\text{swap}} S^n \wedge S^m \simeq S^{m+n}$$

is homotopic to $(-1)^{mn} \text{id}_{S^{m+n}}$.

> Since $\Sigma^\infty : (An, \wedge) \rightarrow (Sp, \otimes)$ is symmetric monoidal, we deduce that for $m, n \geq 0$,

$$S[m+n] \simeq S[m] \otimes S[n] \xrightarrow{\text{swap}} S[n] \otimes S[m] \simeq S[m+n]$$

is $(-1)^{mn} \text{id}$. By shifting, this follows for all $m, n \in \mathbb{Z}$.

Def. A graded ring is an associative ring A together with a decomposition of abelian groups

$$A = \bigoplus_{d \in \mathbb{Z}} A_d$$

Such that

$$(1) A_m A_n \subset A_{m+n}$$

$$(2) \text{ Koszul sign rule: If } a \in A_m \text{ and } b \in A_n, \text{ then } ba = (-1)^{mn} ab.$$

Observe If A is a graded ring, then A_0 is given the structure of a commutative ring. Moreover each graded piece A_d is provided with the structure of an A_0 -module.

Construction. Let A be an E_∞ -ring with unit $u: S \rightarrow A$ and multiplication $\text{mult}: A \otimes A \rightarrow A$. Write

$$\pi_*(A) := \bigoplus_{d \in \mathbb{Z}} \pi_d(A).$$

Define a graded ring structure on $\pi_*(A)$ as follows.

> The unit is given by

$$\mathbb{Z} \simeq \pi_0(S) \xrightarrow{\pi_0(u)} \pi_0(A) \subset \pi_*(A)$$

> The multiplication $\pi_m(A) \times \pi_n(A) \rightarrow \pi_{m+n}(A)$ is given by applying π_0 to

$$(f, g) \longmapsto f \otimes g$$

$$\text{Map}(S[m], A) \times \text{Map}(S[n], A) \longrightarrow \text{Map}(S[m] \otimes S[n], A \otimes A)$$

$$\downarrow \text{mult}_*$$

$$\text{Map}(S[m+n], A)$$

> The Koszul Sign rule Comes from the earlier mentioned fact about the swap map.

> In particular, $\pi_0(A)$ is a commutative ring

Def. A map of E_∞ -rings $f: A \rightarrow B$ is faithfully flat if:

(1) $\pi_0(f): \pi_0(A) \rightarrow \pi_0(B)$ is a faithfully flat map of ordinary rings

(2) For each $n \in \mathbb{Z}$, the induced map of $\pi_0(B)$ -modules

$$\pi_n(A) \otimes_{\pi_0(A)} \pi_0(B) \longrightarrow \pi_n(B)$$

is an isomorphism